

IEEE 5G Core Testbed Installation Guide

Site Deployment & First-Time Setup

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1. Overview

This guide walks you through deploying the IEEE 5G Core Testbed platform on your server. The platform deploys Open5GS 5G Core network functions on a lightweight Kubernetes cluster (k3s), managed through a web-based dashboard. A valid license file (license.jwt) issued by your IEEE coordinator is required.

2. System Requirements

Component	Minimum	Recommended
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
CPU	4 vCPU	8 vCPU
RAM	8 GB	16 GB
Disk	50 GB SSD	100 GB SSD
Network	1 NIC (management)	3 NICs (mgmt + RAN + DN)
Internet	Required (for initial setup)	—

For Multus multi-interface support (N2/N3/N4/N6 separation), your server needs at least 2 physical or virtual NICs. A single NIC setup is supported but routes all traffic through one interface.

3. What You Need Before Starting

1. The deployment package: **ieee_5g_core_deploy_vX.X.tar.gz** (from your IEEE coordinator)
2. Your license file: **license.jwt** (from your IEEE coordinator)
3. Root/sudo access on the target server
4. Internet access from the server (GitHub + container registries)

4. Get Your Cluster UID (for License)

If you don't have a license.jwt yet, complete steps 5.1–5.2 first to install k3s, then send your cluster UID to your IEEE coordinator.

```
# After k3s is installed:
kubectl get namespace kube-system \
  -o jsonpath='{.metadata.uid}'
```

Send this value to your IEEE coordinator to receive your license.jwt.

5. Installation Steps

Step 1 — Transfer files to your server

```
scp ieee_5g_core_deploy_v1.0.tar.gz user@your-server:~
scp license.jwt user@your-server:~
```

Step 2 — Extract the package

```
tar -xzf ieee_5g_core_deploy_v1.0.tar.gz
cd ieee_5g_core_deploy_v1.0
```

Step 3 — Run bootstrap (installs everything)

```
bash bootstrap.sh ~/license.jwt
```

bootstrap.sh will: 1. Verify prerequisites (git, curl, kubectl) 2. Clone the full platform from GitHub 3. Install k3s and system dependencies (~5 minutes) 4. Install your license at /etc/ieee5g/license.jwt 5. Deploy all 5G Core NFs via kubectl apply 6. Wait for all pods to be ready

■ **Warning: Do not interrupt the bootstrap process. If it fails midway, check the error message and re-run. It is safe to run multiple times.**

Step 4 — Verify installation

```
# All pods should show Running
kubectl get pods -n open5gs

# Expected output:
# NAME                READY   STATUS    RESTARTS
# nrf-xxx              1/1     Running   0
# ausf-xxx             1/1     Running   0
# udm-xxx              1/1     Running   0
# amf-xxx              1/1     Running   0
# smf-xxx              1/1     Running   0
# upf-xxx              1/1     Running   0
# core-ui-xxx          1/1     Running   0
```

Step 5 — Access the dashboard

```
kubectl port-forward svc/core-ui-svc 8080:80 -n open5gs
```

Then open your browser at: **http://localhost:8080**

Default credentials:

- Username: admin
- Password: admin123

■ **Warning: Change the default password immediately after first login via Settings → Users.**

6. Network Interface Configuration (Multus)

If your license includes the **multus** feature and you have multiple NICs, configure the network interfaces through the dashboard:

1. Log in to the dashboard
2. Navigate to **Networking** in the left sidebar
3. For each interface card (N2, N3, N4, N6), click the edit icon (pencil)
4. Set the **Host NIC** to your actual interface name (run: `ip link show`)
5. Save — the dashboard will prompt you to restart affected pods

Common NIC names: eth0, eth1, eth2 (older systems) ens3, ens4, ens5 (newer systems) enp3s0, enp4s0 (PCI-named) bond0, bond0.100 (bonded/VLAN)

7. Connecting gNBs

Once the core is running, configure your gNodeBs (srsRAN, OAI, etc.) to connect to the AMF using the parameters below:

Parameter	Value
AMF IP (N2)	192.168.10.10 (Multus) or AMF pod IP
AMF Port	38412
Protocol	SCTP
MCC	001
MNC	01
TAC	1

srsRAN configuration example (gnb.yaml):

```
# gnb.yaml
amf:
  addr: 192.168.10.10
  port: 38412
plmn_list:
  - plmn: "00101"
    tac: 1
```

8. License Status

The license status is shown in the dashboard footer. You will see:

- **Green:** Licensed to {institution} · expires in N days
- **Amber:** ■ License expires in N days — contact your coordinator
- **Red:** License expired — contact your IEEE coordinator immediately

9. Troubleshooting

Symptom	Check	Fix
Pods stuck in Pending	<code>kubectl describe pod</code>	Check node resources, taints
UPF not running	<code>kubectl logs deploy/upf</code>	TUN device, NET_ADMIN permission
AMF not ready	<code>kubectl logs deploy/amf</code>	NRF reachability, config errors
Dashboard shows 403	<code>kubectl logs deploy/core-ui</code>	License file missing or expired
gNB can't connect	AMF logs, firewall	Port 38412/SCTP open, correct IP
License error on login	<code>/api/license endpoint</code>	Check <code>/etc/ieee5g/license.jwt</code>

10. Useful Commands

```
# Check all pods
kubectl get pods -n open5gs

# View logs for a specific NF
kubectl logs -f deploy/amf -n open5gs

# Restart a specific NF
kubectl rollout restart deploy/smf -n open5gs

# Check license status
curl http://localhost:8080/api/license

# Get cluster UID (for license renewal)
kubectl get ns kube-system -o jsonpath='{.metadata.uid}'
```

11. Support

- GitHub Issues: https://github.com/ieee-testbed/ieee_5g_core/issues
- For license issues, contact your IEEE coordinator with your cluster UID
- For deployment issues, include output of: `kubectl get pods -n open5gs && kubectl get events -n open5gs`